COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	09/24/26	**DATE SUBMITTED:**	09/24/26
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ACTIVITY NO. 11.1 Containerization

1. PROCEDURE OUTPUT

Install Docker and enable the docker socket.

Uninstall old versions

Before you can install Docker Engine, you need to uninstall any conflicting packages.

Your Linux distribution may provide unofficial Docker packages, which may conflict with the official packages provided by Docker. You must uninstall these packages before you install the official version of Docker Engine.

The unofficial packages to uninstall are:

- `docker.io`
- `docker-compose`
- `docker-compose-v2`
- `docker-doc`
- `docker-buildx`
- `podman-docker`

Moreover, Docker Engine depends on `containerd` and `runc`. Docker Engine bundles these dependencies as one bundle: `containerd.io`. If you have installed the `containerd` or `runc` previously, uninstall them to avoid conflicts with the versions bundled with Docker Engine.

Run the following command to uninstall all conflicting packages:

```
$ sudo apt remove $(dpkg --get-selections docker.io docker-compose docker-compose-v2 docker-doc docker-buildx
```

sudo apt remove $(dpkg --get-selections docker.io docker-compose docker-compose-v2 docker-doc docker-buildx podman-docker containerd runc | cut -f1)

1. Set up Docker's `apt` repository.

```
# Add Docker's official GPG key:
sudo apt update
sudo apt install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add the repository to Apt sources:
sudo tee /etc/apt/sources.list.d/docker.sources <<EOF
Types: deb
URIs: https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg --print-architecture)
Signed-By: /etc/apt/keyrings/docker.asc
EOF

sudo apt update
```

2. Install the Docker packages.

Latest | Specific version

To install the latest version, run:

```
$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

Note

After installation, verify that Docker is running:

```
$ sudo systemctl status docker
```

If Docker is not running, start it manually:

```
$ sudo systemctl start docker
```

3. Verify that the installation is successful by running the `hello-world` image:

```
$ sudo docker run hello-world
```

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Add to Docker group to your current user.

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ groups
kaizohost adm cdrom sudo dip plugdev users lpadmin docker
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$
```

Create a Dockerfile to install web and DB server.

container > Dockerfile

```
FROM ubuntu:22.04
ENV DEBIAN_FRONTEND=noninteractive
RUN apt-get update && apt-get install -y --no-install-recommends apache2 mariadb-server \
    && rm -rf /var/lib/apt/lists/*
EXPOSE 80 3306
CMD ["bash", "-c", "service mariadb start && apachectl -D FOREGROUND"]
```

Install and build the Dockerfile using Ansible.
playbook.yml

container > playbook-docker.yml

```
- hosts: all
  become: true
  tasks:
    - name: Add Docker repo (CentOS)
      ansible.builtin.get_url:
        url: https://download.docker.com/linux/centos/docker-ce.repo
        dest: /etc/yum.repos.d/docker-ce.repo
      when: ansible_os_family == "RedHat"

    - name: Install Docker (Ubuntu)
      ansible.builtin.apt:
        name: docker.io
        state: present
        update_cache: true
      when: ansible_os_family == "Debian"

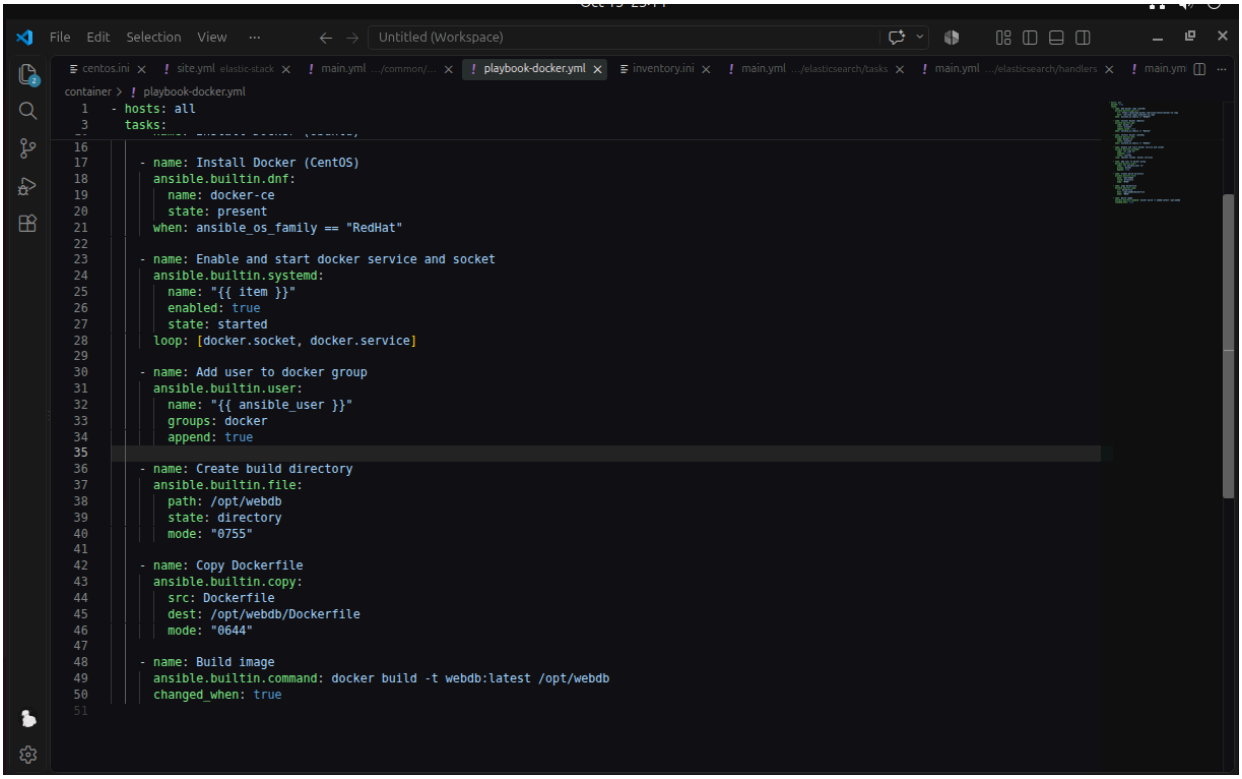
    - name: Install Docker (CentOS)
      ansible.builtin.dnf:
        name: docker-ce
        state: present
      when: ansible_os_family == "RedHat"

    - name: Enable and start docker service and socket
      ansible.builtin.systemd:
        name: "{{ item }}"
        enabled: true
        state: started
      loop: [docker.socket, docker.service]

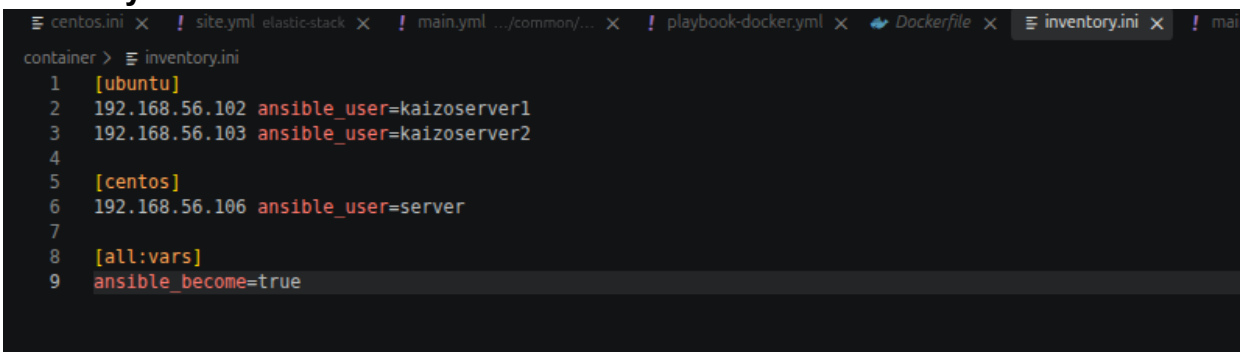
    - name: Add user to docker group
      ansible.builtin.user:
        name: "{{ ansible_user }}"
        groups: docker
        append: true

    - name: Create build directory
      ansible.builtin.file:
        path: /opt/webdb
        state: directory
        mode: "0755"

    - name: Copy Dockerfile
```



inventory.ini



Ubuntu:

kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab/container | kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/container$ ansible-playbook -i inventory.ini playbook-docker
.yml --limit ubuntu -K
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.103]
ok: [192.168.56.102]

TASK [Add Docker repo (CentOS)] ************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

TASK [Install Docker (Ubuntu)] *************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [Install Docker (CentOS)] *************************************************************************************
skipping: [192.168.56.102]
skipping: [192.168.56.103]

TASK [Enable and start docker service and socket] ******************************************************************
ok: [192.168.56.103] => (item=docker.socket)
ok: [192.168.56.102] => (item=docker.socket)
ok: [192.168.56.103] => (item=docker.service)
ok: [192.168.56.102] => (item=docker.service)

TASK [Add user to docker group] ************************************************************************************
ok: [192.168.56.102]
```

kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab/container | kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab

```
skipping: [192.168.56.102]
skipping: [192.168.56.103]

TASK [Enable and start docker service and socket] ******************************************************************
ok: [192.168.56.103] => (item=docker.socket)
ok: [192.168.56.102] => (item=docker.socket)
ok: [192.168.56.103] => (item=docker.service)
ok: [192.168.56.102] => (item=docker.service)

TASK [Add user to docker group] ************************************************************************************
ok: [192.168.56.102]
ok: [192.168.56.103]

TASK [Create build directory] **************************************************************************************
changed: [192.168.56.103]
changed: [192.168.56.102]

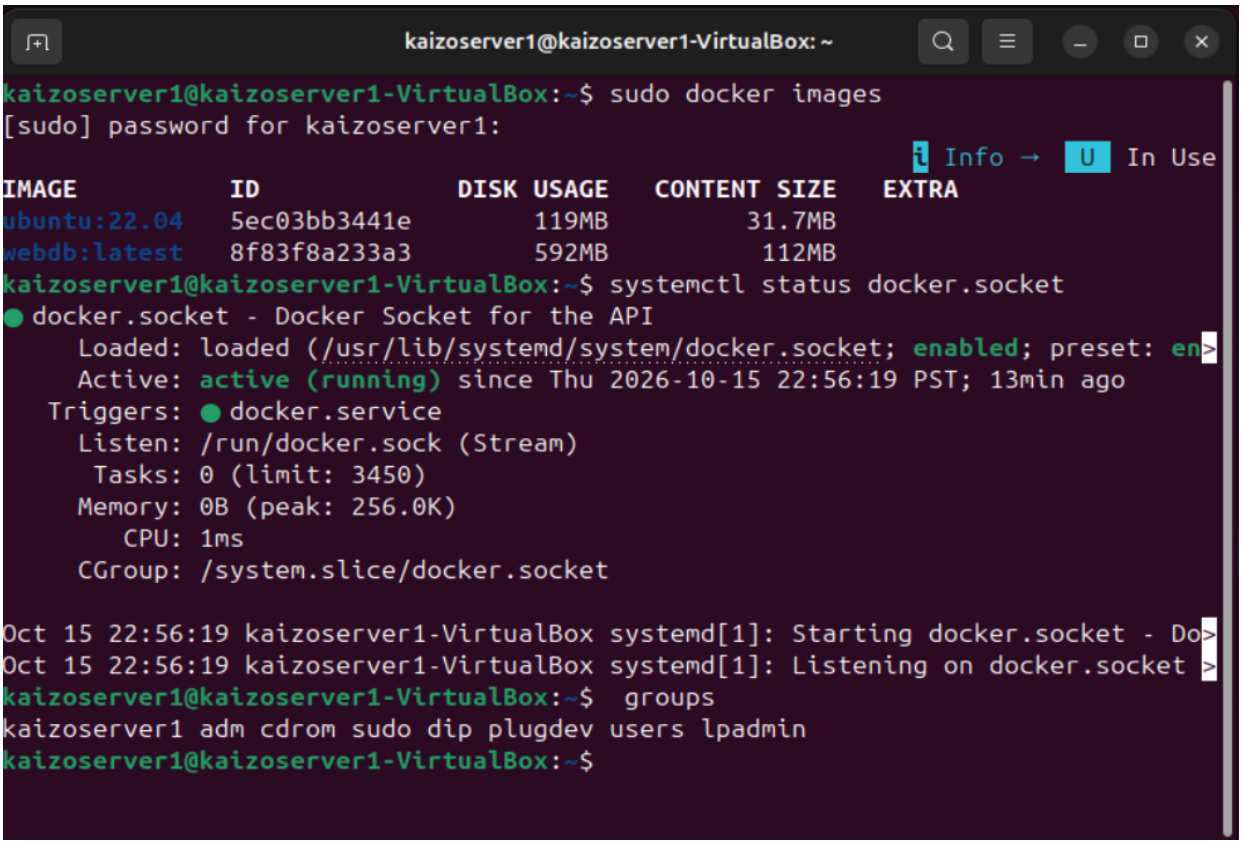
TASK [Copy Dockerfile] *********************************************************************************************
changed: [192.168.56.103]
changed: [192.168.56.102]

TASK [Build image] *************************************************************************************************
changed: [192.168.56.103]
changed: [192.168.56.102]

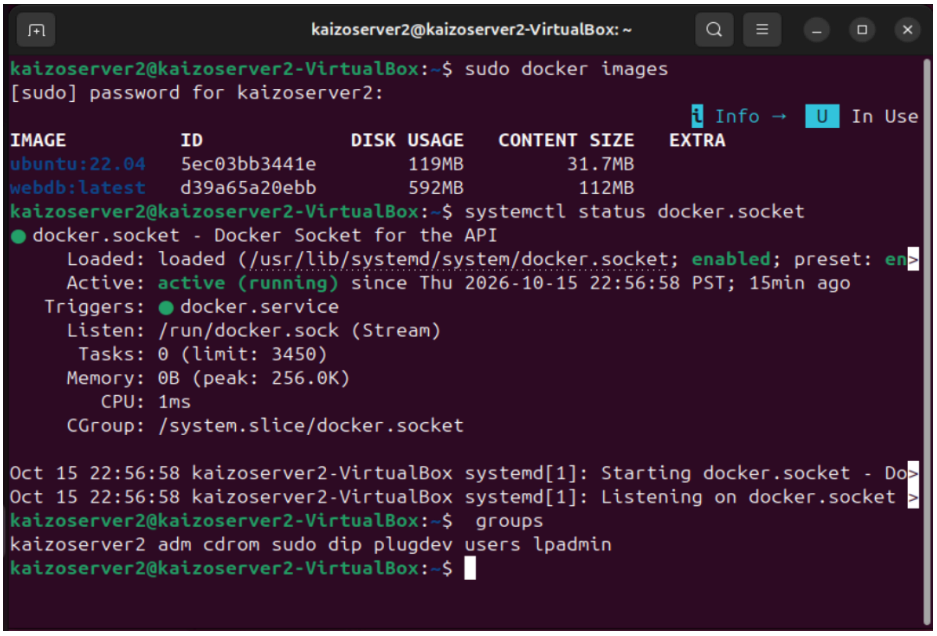
PLAY RECAP *********************************************************************************************************
192.168.56.102             : ok=7    changed=3    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
192.168.56.103             : ok=7    changed=3    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0

kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/container$
```

Server1:



Server2:



Centos:

```
kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab/container   kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab

kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/container$ ansible-playbook -i inventory.ini playbook-docker
.yml --limit centos -K
BECOME password:

PLAY [all] *********************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************
ok: [192.168.56.106]

TASK [Add Docker repo (CentOS)] ************************************************************************************
changed: [192.168.56.106]

TASK [Install Docker (Ubuntu)] *************************************************************************************
skipping: [192.168.56.106]

TASK [Install Docker (CentOS)] *************************************************************************************
changed: [192.168.56.106]

TASK [Enable and start docker service and socket] ******************************************************************
changed: [192.168.56.106] => (item=docker.socket)
changed: [192.168.56.106] => (item=docker.service)

TASK [Add user to docker group] ************************************************************************************
changed: [192.168.56.106]

TASK [Create build directory] **************************************************************************************
changed: [192.168.56.106]

TASK [Copy Dockerfile] *********************************************************************************************
changed: [192.168.56.106]
```

```
kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab/container   kaizohost@kaizohost-VirtualBox: ~/CPE212_SanJuan/ansible-lab

TASK [Add Docker repo (CentOS)] ************************************************************************************
changed: [192.168.56.106]

TASK [Install Docker (Ubuntu)] *************************************************************************************
skipping: [192.168.56.106]

TASK [Install Docker (CentOS)] *************************************************************************************
changed: [192.168.56.106]

TASK [Enable and start docker service and socket] ******************************************************************
changed: [192.168.56.106] => (item=docker.socket)
changed: [192.168.56.106] => (item=docker.service)

TASK [Add user to docker group] ************************************************************************************
changed: [192.168.56.106]

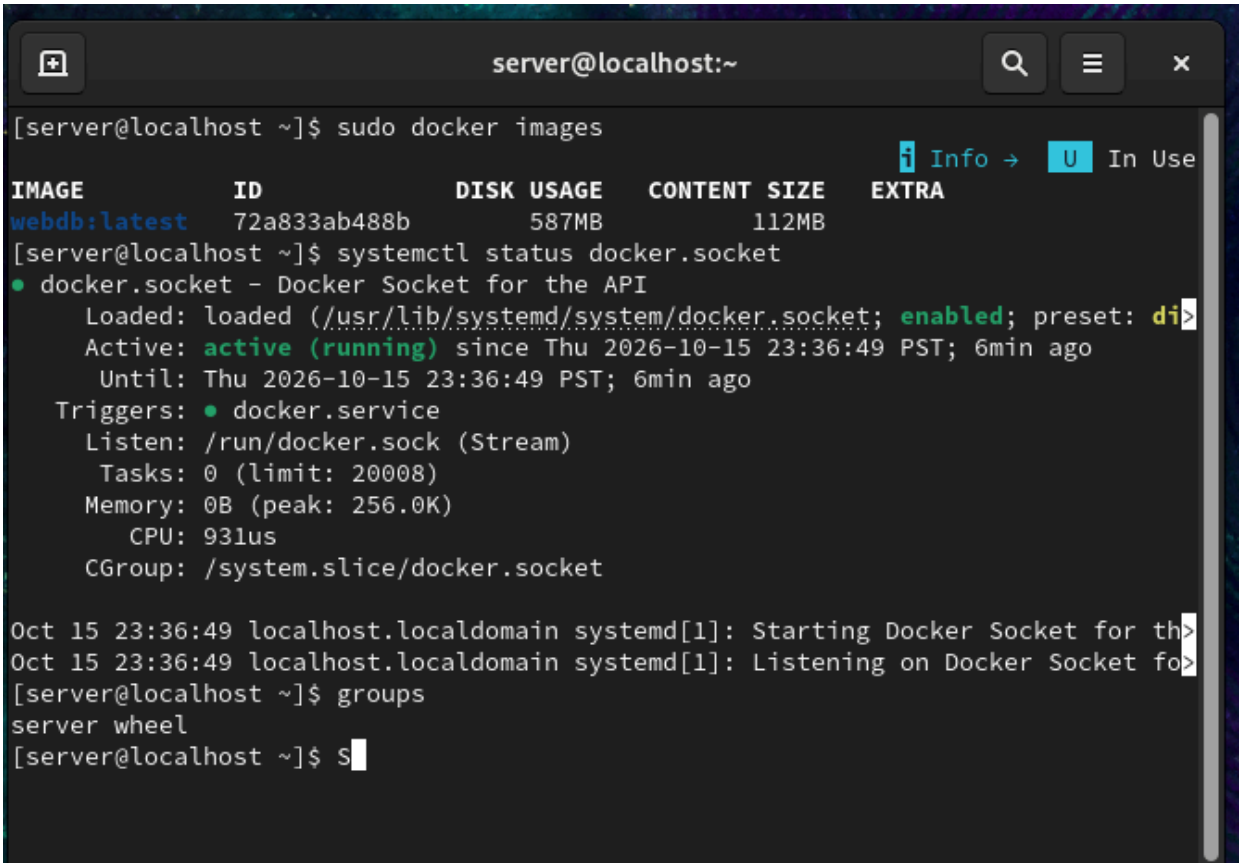
TASK [Create build directory] **************************************************************************************
changed: [192.168.56.106]

TASK [Copy Dockerfile] *********************************************************************************************
changed: [192.168.56.106]

TASK [Build image] *************************************************************************************************
changed: [192.168.56.106]

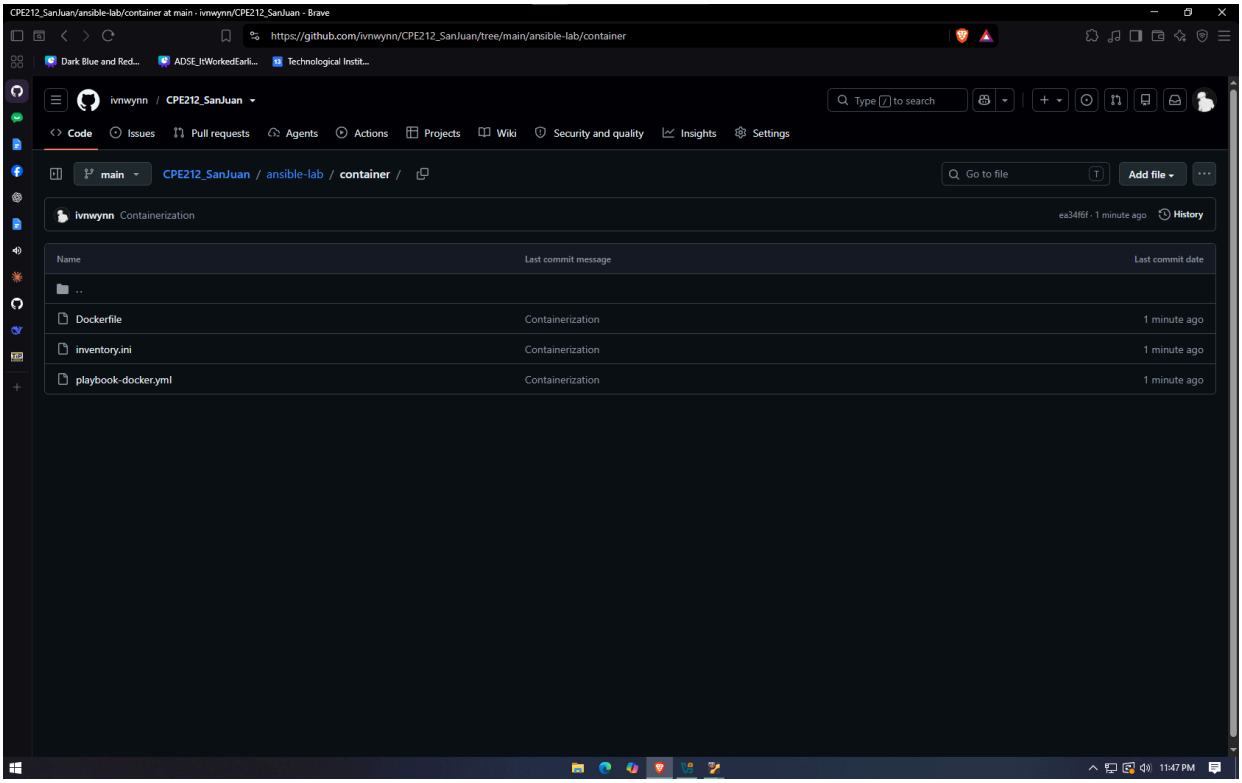
PLAY RECAP *********************************************************************************************************
192.168.56.106             : ok=8    changed=7    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0

kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab/container$
```



Add, commit and push it to your repository.

```
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git add .
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git commit -m "Containerization"
[main ea34f6f] Containerization
 6 files changed, 65 insertions(+), 37 deletions(-)
 create mode 100644 ansible-lab/container/Dockerfile
 delete mode 100644 ansible-lab/container/ansible-draft.yml
 delete mode 100644 ansible-lab/container/inventories/centos.ini
 delete mode 100644 ansible-lab/container/inventories/ubuntu.ini
 create mode 100644 ansible-lab/container/inventory.ini
 create mode 100644 ansible-lab/container/playbook-docker.yml
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$ git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 3 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.25 KiB | 1.25 MiB/s, done.
Total 7 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ivnwynn/CPE212_SanJuan
   eaa235e..ea34f6f  main -> main
kaizohost@kaizohost-VirtualBox:~/CPE212_SanJuan/ansible-lab$
```



CONCLUSION:

In this activity, I used Ansible to automate the setup of a containerized web and database server across multiple Ubuntu servers. The playbook installed Docker, enabled the Docker socket, added the user to the docker group, and built an image from a Dockerfile containing Apache and MariaDB. I ran into a few errors along the way, such as a missing Dockerfile and a missing destination directory, and fixing them taught me that Ansible runs tasks in order and that the copy module does not create directories on its own. Overall, the activity showed how automation makes server setup faster, consistent, and repeatable, and how Git keeps the work versioned and easy to share.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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